

OFFLINE LOCAL KNOWLEDGE

The phone does not need the biggest model. The mesh needs the better memory.

A system architecture for turning repurposed phones into an offline knowledge network where local facts are signed, witnessed, refined and passed device to device.

The problem

On-device models are being asked to know the world alone. A phone has limited compute, limited storage and a model whose local knowledge may already be stale.

The failure is architectural, not only model-sized.

The Helius answer

Move memory and verification into the network. Phones exchange compact, signed knowledge packets over a peer-to-peer mesh. A small local model retrieves and explains them.

“Best vegan restaurants in the city I am currently in” is not a benchmark that demands a larger model. It demands recent, local, witnessed evidence.

Observe

A person, place or sensor creates a local claim.

Witness

Independent evidence and signatures add confidence.

Gossip

Nearby phones carry the packet without the internet.

Answer

The local model retrieves, ranks and explains.

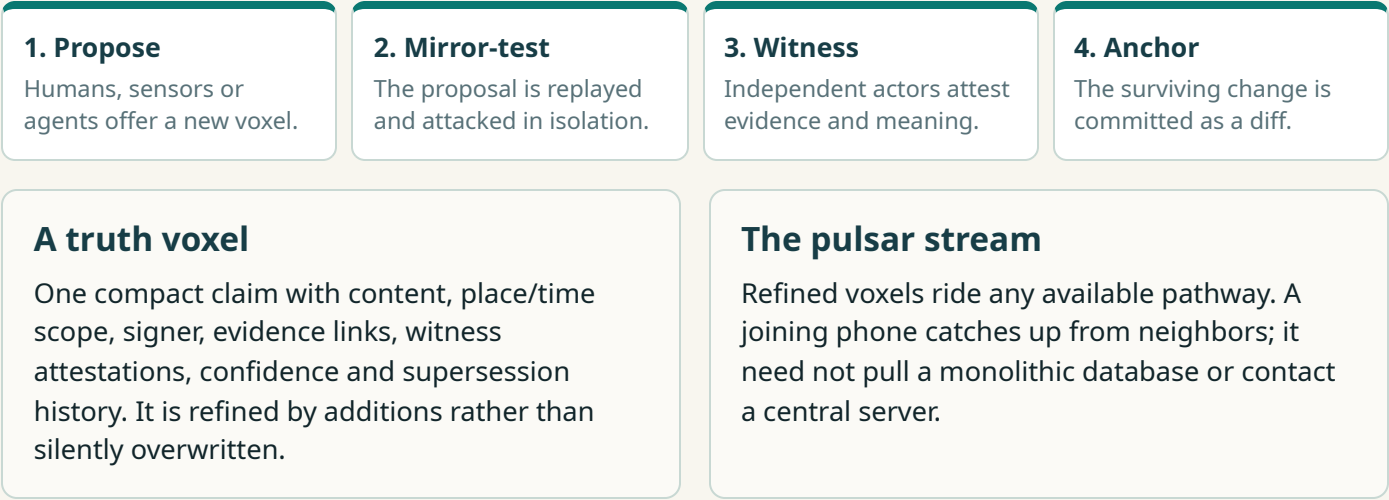
Status: the end-to-end architecture is documented. The offline knowledge application is not built yet. This handoff separates the designed system from the implementation still required.

Repurposed phones become a distributed knowledge fabric

Helius starts with wiped phones running a preloaded operating system. Each device is a peer: it can discover neighbors, store-and-forward messages, verify signed content, run a small local model and recover a known-good state. The blockchain is an audit rail, not the knowledge database.

L6	Interface	Query, voice, generated views, human-readable provenance
L5	Compute	Local inference, task market, optional dock or “whale” compute
L4	Agentic fabric	Intent routing, translation, retrieval and explanation
L3	Identity and routes	Device identities, wallets, route selection and incentives
L2	Mesh transport	Bluetooth / Wi-Fi, peer discovery, delay-tolerant store-and-forward
L1	Kernel and link	Custom Android base, radio control, verified boot pathways
L0	Silicon and trust root	Repurposed phones, hardware keystores, radios and optional docks

Knowledge moves through a four-stage refinement loop



Diff-chain: only changes are anchored: previous hash + diff + new hash. Content remains distributed; the chain preserves ordering, integrity and reconstructability.

The hard travel query becomes a local evidence query

The first demo should not compete with a laptop model on general knowledge. It should prove something a laptop model cannot guarantee: fresh, local answers assembled from nearby human evidence while offline.

Tell me the best vegan restaurants in the city I am currently in.

01 Discover nearby knowledge

The phone asks peers for vegan-restaurant voxels within a chosen radius. Queries and replies travel by Bluetooth or Wi-Fi store-and-forward.

02 Filter what can be checked

Candidate packets expose when and where the observation was made, who signed it, what evidence was attached and how many independent witnesses confirmed it.

03 Rank for the actual user

The local model combines freshness, distance, dietary fit, witness diversity and the user's preferences. It does not invent missing hours, menus or availability.

04 Answer with provenance

The response names the best-supported options and shows why: "three recent witnesses," "menu photo from yesterday," or "conflicting report; verify before traveling."

POIDH-STYLE BOUNTY LOOP

Bounty — Evidence — Witnesses — Voxel — Offline answer

A "pics or it didn't happen" bounty already supplies the incentive to gather evidence. Helius does not replace that mechanism; it wraps it with offline transport, independent witnessing, refinement and durable peer-to-peer availability.

What the prototype should measure

Recall

Can an offline phone find all nearby witnessed entries?

Freshness

How quickly does a new or corrected voxel propagate?

Trust

Does ranking improve when witnesses disagree or collude?

Resilience

Does knowledge survive disconnection and node loss?

The architecture is complete enough to build the smallest proof

Documented

- Repurposed-phone mesh and custom operating-system vision
- Layered L0-L6 system model
- Truth-voxel data primitive and refinement cycle
- Pulsar-style peer gossip and catch-up
- Signed containers and diff-only anchoring
- Mirror security plane and checkpoint rollback
- Genesis baseline, provenance and append-only sublines

Not built or settled

- Working offline local-knowledge application
- Final radio and mesh protocol selection
- Device attestation and Sybil resistance
- Production incentive and token economics
- Custom ROM, packaging and update system
- Field performance, abuse resistance and privacy validation

The Mirror keeps experiments off the live system

Every signed container and diff is replayed in an isolated virtual machine against an agreed baseline. Suspicious behavior receives false paths and decoy inputs. A watcher records the behavior; a poisoned container can be quarantined and the phone restored to its last clean checkpoint. These are established security techniques composed into one phone-shaped control plane; the composition is the proposal.

FIRST PROOF Two or three phones. One neighborhood. One query family.

1. Load a small signed catalog of local vegan venues onto one device.
2. Create a bounty-backed observation with photo evidence and a second witness.
3. Gossip the voxel across disconnected phones using store-and-forward.
4. Ask the same location query on a phone with no internet and no prior catalog.
5. Correct or revoke one detail, then verify that the diff propagates and the old state remains auditable.

COLLABORATION OFFER

Walk through the architecture together, or build this narrow demo together. The useful comparison is not phone model versus laptop model. It is isolated phone versus witnessed local mesh.

Architecture basis: Helius Network Vision; Helius Phone Layers 0-6 + the Mirror; The Truth Voxel Stream; The Genesis Mother Block. Design package dated 25 September 2026.